DDS Function Generator

TFG3200E Series

Features

- ✓ Max. output frequency 5MHz/10MHz/15MHz/20MHz
- ✓ 2 output channels
- ✓ Mono LCD display
- ✓ Direct Digital Synthesis technology (DDS)
- ✓ Min. output amplitude 1mV (50Ω) with good stability
- ✓ Sampling rate 100MSa/s, vertical resolution 8 bits, waveform length 1024 points
- ✓ 32 built-in waveforms
- ✓ 40 sets save & recall for panel settings
- ✓ Modulations: FM, FSK, ASK, PSK
- ✓ Frequency sweep, amplitude sweep, burst and TTL output functions
- ✓ Over voltage, over current, short circuit and reverse voltage protections
- ✓ High speed rotary dial and keypad input
- ✓ Standard 200MHz external frequency counter
- ✓ Optional RS-232 interface for PC remote control
- ✓ Optional power amplifier

Product photo





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Specifications

Model		TFG-3205E	TFG-3210E	TFG-3215E	TFG-3220E
Output frequency		**1μHz~5MHz**	**1μHz~10MHz**	**1μHz~15MHz**	**1μHz~20MHz**
Waveform					
Output waveform		32 built-in waveforms, including Sine, Square, Triangle, Ramp, Pulse, etc.			
Waveform length		1024 points			
Vertical resolution		8 bits			
Sampling rate		100MSa/s			
Sine	Harmonic distortion	≥40dBc (<1MHz); ≥35dBc (1~20MHz)			
	Total distortion	≤1% (20Hz~200kHz)			
Square	Rise/fall time	≤35ns			
	Overshoot	≤10%			
	Duty cycle	1%~99%			
Frequency					
Range	Sine	1μHz~5MHz	1μHz~10MHz	1μHz~15MHz	1μHz~20MHz
	Square	1μHz~5MHz			
	Other	1μHz~1MHz			
Resolution		1μHz			
Accuracy		$\pm 5 \times 10^{-5}$			
Stability		$\pm 5 \times 10^{-6}$ /3hours			
Output characteristics					
Amplitude	Range	2mVpp~20Vpp (open circuit, ≤10MHz)			
		2mVpp~15Vpp (open circuit, 10MHz~15MHz)			
		2mVpp~8Vpp (open circuit, 15MHz~20MHz)			
	Resolution	20mVpp (amplitude>2Vpp); 2mVpp (amplitude<2Vpp)			
	Accuracy	±(1%+2mVrms) (open circuit, 1kHz, sine)			
	Stability	±0.5% /3hours			
	Flatness	±5% (<10MHz); ±10% (>10MHz)			
	Output impedance	50Ω			
Offset	Range	±10V (open circuit, attenuation 0 dB)			
	Resolution	20mVdc			
	Accuracy	±(1%+20mVdc)			
Sweep					
Parameter		Frequency, Amplitude			
Range		Free to set start and stop point			
Time		100ms~900s			
Direction		Up, Down, Up-Down			
Mode		Linearity, Logarithmic			
Control		Auto sweep or manual sweep			
Frequency Modulation (FM)					
Carrier signal		CHA signal			
Modulating signal		CHB or external signal			
Deviation		0%~20%			
Shift Keying					
FSK		Free to set the hop frequency and the carrier frequency			
ASK		Free to set the hop amplitude and the carrier amplitude			
PSK		Hop phase: 0~360°, resolution: 1°			
Alternative rate		10ms~60s			

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Burst	
Carrier signal	CHA signal
Trigger signal	TTL_A signal
Burst counts	1~65000 cycles
Trigger source	Internal TTL, External, Single
CHB output characteristics	
Output waveform	32 built-in waveforms, including Sine, Square, Triangle, Ramp, Pulse, etc.
Waveform length	1024 points
Vertical resolution	8 bits
Sampling rate	12.5MSa/s
Frequency range	Sine: 1μHz~1MHz; Other: 1μHz~100kHz
Frequency resolution	1μHz
Frequency accuracy	$\pm 1 \times 10^{-5}$
Amplitude range	50mVpp~20Vpp (open circuit)
Amplitude resolution	20mVpp
Output impedance	50Ω
CHB signal is used as burst signal	
Carrier signal	CHB signal
Trigger signal	TTL_B signal
Burst counts	1~65000 cycles
Trigger source	Internal TTL, External, Single
TTL output	
Waveform	Square, rise/fall time ≤20ns
Frequency	10mHz~1MHz
Amplitude	TTL, CMOS compatible, low<0.3V, high>4V
Frequency counter	
Frequency range	1Hz~200MHz
Input amplitude	100mVpp~20Vpp
Power amplifier (optional)	
Max. output power	7W (8Ω), 1W (50Ω)
Max. output voltage	22Vpp
Frequency bandwidth	1Hz~200kHz
General	
Operation characteristics	Key operation for all functions, Menu display, Rotary dial adjustment
Display	Mono LCD
Language	English, Chinese (simplified), Chinese (traditional)
Interface	Optional RS-232 interface
Operating environment	0~40℃, <80%RH
Power source	AC110V/220V±10% selectable, 50/60Hz, Max. 45VA
Standard accessories	Power cord x1, Operation manual x1, BNC-BNC cable x1, Test lead x1
Optional accessories	Software CD x1, RS-232 cable x1
Dimension (WxHxD)	260x110x385mm
Weight	3.5kg

Specifications are subject to change without prior notice.

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